Venkata Lakshmi Parimala Pasupuleti

Senior SWE · AI Platform · LLM-Powered Features · Agents · RAG · Evals · MCP · LLM Security · AWS · Python

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SUMMARY

Senior software engineer building AI platform infrastructure end-to-end - agents, retrieval systems, evaluation harnesses, and LLM security. Currently at Intuit integrating Anthropic Claude into production financial services: structured outputs, prompt design, retrieval grounding, LLM-as-judge eval regression tests, guardrails, and AWS CI/CD. Built **Leasel**: 8-agent LangGraph platform with tool/function calling, hybrid search RAG pipeline (embeddings, chunking, semantic reranking), MCP server integration, human-in-the-loop controls, and prompt injection defenses enforced in code. Built **AgentProbe**: open-source LLM evaluation toolkit with offline golden datasets, online metrics, LLM-as-judge scoring, cost-aware multi-provider routing (Anthropic, OpenAI, Azure OpenAI, AWS Bedrock, Groq, Ollama), and CI regression harness - the eval infrastructure that gates releases and tracks quality over time. At Entrata (7+ years, 100K+ users), owned production systems at scale in a regulated industry. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

LLM-Powered Features · Agents · Tool Calling | Prompt Design · Structured Outputs · Prompt Caching | RAG Pipelines · Hybrid Search · Chunking · Ranking

Embeddings · Vector Search · Retrieval Systems | LLM Evaluation · Offline Evals · LLM-as-Judge | Golden Datasets · CI Regression Harness

MCP Server · Agent Frameworks · LangGraph | LLM Security · Prompt Injection Defense · Red-Teaming | Permission Models · Least-Privilege · Audit Logging

Cost Optimization · Model Routing · AWS Bedrock | Tracing · Observability · Monitoring | Python · TypeScript · Node.js · SQL

AWS · Docker · Kubernetes · GitHub Actions CI/CD | Regulated Industry · HIPAA/PHI · Compliance | Mentoring · Code Review · System Design

EXPERIENCE

Software Engineer

Jun 2026 - Present

Intuit — Virginia, USA

- **Integrate Anthropic Claude into production Python/FastAPI services** - financial data summaries, natural-language queries, and reporting; design prompts and structured-output schemas so answers stay consistent and are traceable to source records via retrieval grounding.
- **Build LLM evaluation and regression tests** - LLM-as-judge scoring and golden datasets to catch hallucinations and quality drops before release; guardrails around LLM calls with input validation, sensitive-data handling, and fallbacks when model output fails checks.
- Build ML pipelines (scikit-learn, XGBoost) from data ingestion and feature engineering to prediction APIs; develop React/TypeScript front ends on AWS; containerize services with Docker through GitHub Actions CI/CD with logging and monitoring for production visibility.

Senior Software Engineer

Oct 2020 - Jul 2024

Entrata — Apartment & Affordable Housing SaaS (regulated industry, 100K+ monthly active users)

- Delivered 20+ features end-to-end (Python, PHP, Java, JavaScript, React, PostgreSQL) across 120+ Agile sprints; maintained HUD income-limit import pipeline for 6 years (federal data → ~180,000 records with validation and rollback each cycle).
- Built RabbitMQ-based async processing system for rent updates across thousands of tenant records; Python/Bash automation cutting manual operational work by 80%; internal debugging tools reducing triage time by 40%.
- Optimized 200+ SQL queries (30% performance gain); Docker microservices redesign (50% better scalability); AWS S3 migration with Libsodium PHI encryption (100% coverage, zero incidents); reviewed 300+ pull requests and mentored junior engineers.

Junior Software Engineer

Sep 2018 - Oct 2020

Entrata India — Pune

- Built property management and compliance modules (Python, PHP, Java, PostgreSQL); delivered large-scale data import workflows (1.6M records); led AWS S3 cloud migration from legacy storage.

PROJECTS

Leasel - AI Platform: Agents · RAG · MCP · LLM Security 2026 | Python, LangGraph (8-agent), FastAPI, ChromaDB, MCP, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production AI platform with three surfaces sharing one retrieval layer: **8-agent LangGraph workflow** with tool/function calling, durable state, and conditional routing; **RAG pipeline** with OCR ingestion, section-aware chunking, hybrid search (TF-IDF + semantic reranking), embeddings, and citation grounding; **MCP server integration** for shared tool access; **LLM security throughout** - prompt injection defenses, least-privilege permission model, access-controlled retrieval, audit logging on every agent action.
- CI regression harness; 93.6% precision and 87.9% F1 across 1,240 test cases; human-in-the-loop escalation gates all regulated decisions.

AgentProbe - LLM Eval Infrastructure & Cost-Aware Multi-Provider Routing 2025 | Python, FastAPI, LangGraph, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- Open-source eval toolkit: **offline evals** with golden datasets and LLM-as-judge scoring; **online metrics** tracking quality over time; CI regression harness that gates releases on eval thresholds; **cost-aware model routing** across Anthropic, OpenAI, Azure OpenAI, AWS Bedrock, Groq, and Ollama - routes work to cheapest model that meets quality bar; 5-metric eval engine (accuracy, hallucination rate, retrieval quality, completeness, compliance); tracing and structured logging across all agent runs.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia | GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017